QUASI-COMPACT AND DOMINANT IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the author's parenthetical noting that the surjective morphism used in the descent direction need not be flat; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 2 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Dominant morphisms are preserved by flat base change. Morphisms, Lemma morphisms-lemma-base-change-dominant, invoked at proof lines 5842-5844. Gives the easy direction of the locality.
- Quasi-compactness is fpqc local on the base. Same chapter, Lemma lemma-descending-property-quasi-compact, invoked at proof lines 5845-5846. Supplies the quasi-compact half of the property in the descent direction.
- A composite of two dominant morphisms is dominant, and dominance can be tested after composing with a surjective morphism. Standard, used at proof lines 5848-5853. Gives dominance of the original morphism from dominance of its base change.

Source reference. Original lemma 02KQ: sources/stacks-descent/pinned-original-source.tex, source lines 5688–5688. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0H3F: sources/stacks-chow/tags.txt, source lines 21315–21315. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$ "f is quasi-compact and dominant" is fpqc local on the base.*

Proof. By Morphisms, Lemma 0H3F, quasi-compact dominant morphisms are preserved by flat pullback. The other direction is easier. Indeed, quasi-compactness is fpqc local on the base by Lemma 02KQ. Being dominant is clearly Zariski local on

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the base. Finally, let $S' \to S$ be a surjective morphism of schemes (not necessarily flat), and let $f : X \to S$ be a morphism. Assume that the base change $f' : X' \to S'$ is dominant. Then $X' \to S' \to S$ is a composite of two dominant morphisms, hence dominant. Since this is also the composite $X' \to X \to S$, it follows that $X \to S$ is dominant. $\square$